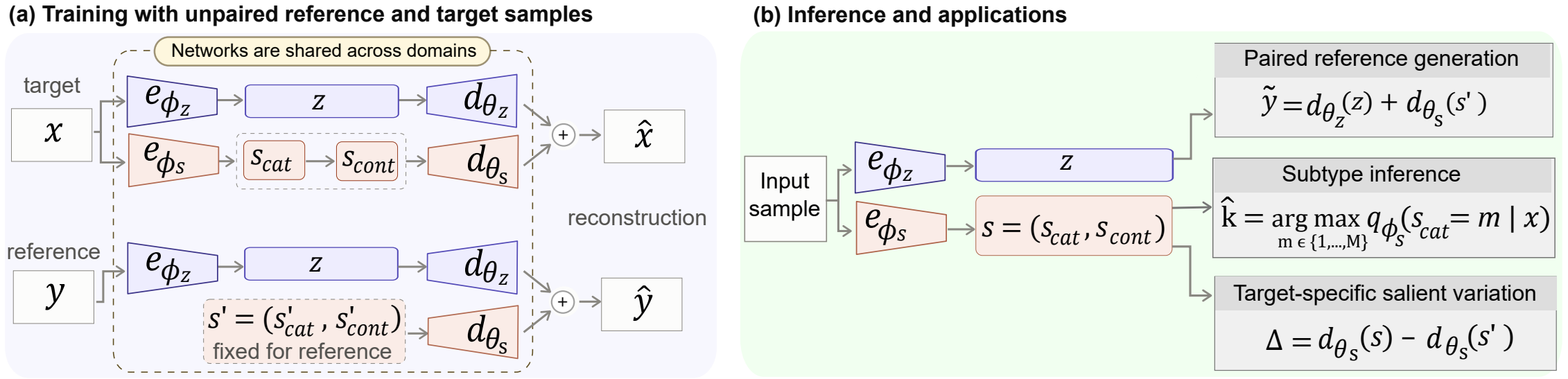

(a) Training with unpaired reference and target samples
Networks are shared across domains
target
x
e_{ϕ_z}
z
d_{θ_z}
e_{ϕ_s}
s_{cat}
s_{cont}
d_{θ_s}
$\hat{x}$
reconstruction
reference
y
e_{ϕ_z}
z
d_{θ_z}
$s' = (s'_{cat}, s'_{cont})$
fixed for reference
d_{θ_s}
$\hat{y}$
(b) Inference and applications
Input sample
e_{ϕ_z}
z
e_{ϕ_s}
$s = (s_{cat}, s_{cont})$
Paired reference generation
$\tilde{y} = d_{\theta_z}(z) + d_{\theta_s}(s')$
Subtype inference
$\hat{k} = \arg\max_{m \in \{1,\ldots,M\}} q_{\phi_s}(s_{cat} = m \mid x)$
Target-specific salient variation
$\Delta = d_{\theta_s}(s) - d_{\theta_s}(s')$